for this proposition was a relatively junior rocket enthusiast, Wernher von Braun. He was working for the army by January 1933, when Adolf Hitler became Chancellor of Germany.

The development of a large rocket capable of striking distant targets attracted ever-increasing resources from the Nazi state. By 1936 von Braun had 80 workers at his command and a top secret research and development facility was under construction at Peenemünde in a remote area on the Baltic. It eventually included a rocket factory, launch pads, and a liquid oxygen plant. By 1943 some 6,000 scientists, technicians, and engineers were employed there, along with countless forced laborers and prisoners of war.

The product of their efforts was the A-4, a liquid-propellant missile that, when fired vertically, reached an altitude of 110 miles (176km). Renamed by Nazi propaganda chief Joseph Goebbels as the V-2 (V for *Vergeltungswaffe*, or "vengeance weapon"), it was used against the Allies from the summer of 1944. It fell far short of the decisive impact the Nazis had hoped for – each V-2 launched against Britain killed on average 1.76 people – but it revealed the existence of a revolutionary new technology to the Western Allies and the Soviet Union. Getting their hands on that technology was a major priority for both in 1945.

> "A failed weapons system, the product of horrific slave labor, the V-2... opened the door to the universe."
>
> **Tom D. Crouch**
> AIMING FOR THE STARS

Von Braun's position, as Germany's defeat loomed, was on the face of it a desperate one. He was a member of the SS and his secret weapon had been produced by slave labor under murderous conditions. But he knew that the knowledge he and his team possessed was a powerful bargaining counter. By the end of 1945, along with captured V-2s, the Americans had carried von Braun and some of his colleagues back to the United States to work for their former enemies as "prisoners of peace." Over the following years about 120 of the 6,000 from Peenemünde reassembled, eventually creating an expatriate colony in Huntsville, Alabama. Their first collective task was to develop America's first nuclear ballistic missile, the Redstone, which was test launched from Cape Canaveral, Florida, in 1953.

FILMING THE LAUNCH
Recorded by movie cameras, a test launch of the Bumper rocket, using a captured V-2 as first stage, takes place at Cape Canaveral on July 24, 1950. Although rocket enthusiasts were inspired by the idea of space exploration, funding for rocket research depended on its military potential.

WERNHER VON BRAUN

Born into the Prussian landed aristocracy, Wernher von Braun (1912–77) was caught up in the "rocket craze" of the 1920s and in 1932 agreed to work for the German army. He always claimed that he had no interest in weapons research, but was "milking the military purse" to develop rockets that would eventually travel into space. After heading the Nazi program that produced the V-2 rocket, at the end of World War II he engineered the transfer of his key staff to the United States. There he emerged in the 1950s as a high-profile advocate of travel to the Moon and Mars, fronting Disney TV shows and writing influential articles for Collier's magazine. His team played a key role in the development of US ballistic missiles and of launch rockets for the space program.

CLEAN IMAGE
His past in Nazi Germany was rarely allowed to cast a shadow over von Braun's reputation as an inspired rocket pioneer.